

# An Nguyen

hongan.nguyen04@gmail.com — an-h-nguyen.com — github.com/Noctural123 — linkedin.com/in/annguyen123

## EDUCATION

### University of Oklahoma

Norman, OK

M.S. in Computer Science — GPA: 3.9

Expected: Fall 2026

B.S. in Computer Science & B.A. in Mathematics — GPA: 3.77

Graduated: Fall 2025

## SKILLS

**Languages:** Python, Java, TypeScript, JavaScript, C#, C++, PHP, SQL

**Web & Frontend:** React, Next.js, TypeScript, HTML, CSS, Chrome Extensions (MV3)

**Backend & Data:** Node.js, Express, .NET, REST APIs, microservices, PostgreSQL, MySQL, MongoDB, Prisma

**AI & LLM:** Claude, ElevenLabs & fal.ai APIs, prompt engineering, scikit-learn, Claude Code, Cursor

**Cloud & Tools:** AWS, GCP, Docker, Git, GitHub Actions, CI/CD, automated testing, observability

## EXPERIENCE

### Paycom

Oklahoma City, OK

*Software Development Intern (PHP, SQL)*

May 2025 – August 2025

- Standardized structured logging across **12+** distributed microservices, improving observability and tracing.
- Reduced noisy error log volume by **38%** via a MySQL-backed toggle/TTL system, speeding incident triage.
- Built an admin UI to toggle Monolog processors at runtime, enabling targeted debugging without redeloys.

*Software Development Intern (C#, .NET, SQL)*

May 2024 – August 2024

- Built a vendor API emulator logging **100%** of third-party traffic to MySQL for deterministic replay and tests.
- Shipped a React/TypeScript investigation console with PM/QA teams, cutting cross-team triage time by **~40%**.
- Achieved **~80%** coverage of third-party screening API calls, enabling real vendor behavior replay.

### U.S. Department of Defense – Tinker Air Force Base

Oklahoma City, OK

*Software Engineering Intern (Java)*

May 2023 – November 2023

- Built a Java maintenance app serving **15+** B-2 engineering teams, cutting end-to-end workflow time.
- Implemented form validations, cutting invalid submissions by **~35%** and reducing manual review load.
- Grew automated test coverage from **~45%** to **~75%** via unit/integration tests and CI improvements.

### OU – Gallogly College of Engineering – Undergraduate Research

Norman, OK

*Undergraduate Researcher*

August 2023 – January 2024

- Conducted a systematic literature review on AI/ML in racing games; presented at the ASEE conference.
- Synthesized **20+** peer-reviewed studies via multi-database search with inclusion/exclusion criteria.
- Worked with a faculty mentor to validate methodology and produce publication-ready findings.

## PROJECTS

### Quero – AI Video Generation SaaS — Next.js, React, TypeScript, PostgreSQL, Stripe, FFmpeg

2026

- Built a full-stack AI SaaS with Claude Code and Higgsfield: one prompt in, a publishable video out.
- Streamlined social media content creation via a brief-to-storyboard flow, scene voiceovers, and in-app editing.
- Designed a crash-safe LLM pipeline across **4** AI APIs that resumes mid-run from any failed stage.
- Engineered an append-only Stripe credit ledger with transactional debits, preventing charges for failed runs.

### F1 Telemetry – CAN Bus Intrusion Detection — Python, scikit-learn, FastF1, pandas, Arcade

2026

- Simulated in-vehicle CAN bus traffic and attacks (injection, fuzzing) from **2.8M+** real F1 telemetry points.
- Built a **3-layer** hybrid ML intrusion detection system (Isolation Forest, Random Forest) catching **100%** of attacks.
- Exposed inflated single-run ML accuracy via cross-run evaluation, training on **4** runs and testing on unseen data.

### LeetCode Recommendation Engine — Python, React, TypeScript, Node.js, PostgreSQL, Prisma, AWS

2025

- Designed a Python ranking algorithm recommending each user's next problem from mastery and difficulty gaps.
- Shipped a Chrome extension (MV3) and React/TypeScript dashboard tracking **150** problems, backed by AWS.
- Accelerated delivery with AI dev tools (Claude Code, Cursor), validating all generated code via review and tests.

## ACTIVITIES AND LEADERSHIP

### Vietnamese Eucharistic Youth Movement

Oklahoma City, OK

*Youth Leader, Lion Dancer*

Aug 2014 – Present

- Cultivate faith, leadership, and responsibility among **400+** children through cultural education.
- Featured lion dancer at an OKC Thunder NBA game; fundraise annually via Lunar New Year performances.
- Collaborate with youth leaders to plan seasonal camps, weekly lessons, and team-building activities.